

Program : Diploma in Renewable Energy	
Course Code : 4298	Course Title:- Renewable Energy Technology Lab
Semester : 4	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To understand the basic concepts and working of Wind energy conversion system
- To impart the basic knowledge of the production of energy from Biomass
- To illustrate the basic working principle of a fuel cell
- To understand the working of a micro hydro turbine energy generator

Course Prerequisites:

Topic	Course Code	Course Name	Semester
		Basic Renewable Energy workshop	III

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the operation of a wind power generator	12	Applying
CO2	Demonstrate the construction of Biogas plants	6	Applying
CO3	Identify the working of a fuel cell and measure its parameters	9	Applying
CO4	Identify the working of a micro Hydro Turbine Generator	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3	3		
CO2				3	3		
CO3				3	3		
CO4				3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Identify the operation of a wind power generator		
M1.01	Study different components of a Wind power Plant .	3	Understanding
M1.02	Measure wind Speed using an anemometer	3	Applying
M1.03	Plot the power and torque characteristics of a wind turbine .	3	Applying
M1.04	Plot I-V characteristics of wind turbines at different wind speeds.	3	Applying
CO2	Demonstrate the construction of Biogas plants		
M2.01	Show the construction of fixed dome type biogas plant	3	Understanding
M2.02	Demonstrate the construction of drum type biogas plant	3	Understanding

	Lab Test	3	
CO3	Identify the working of a fuel cell and measure its parameters		
M3.01	Identify various parts a fuel cell	3	Understanding
M3.02	Measure the output voltage and output current of a fuel cell	3	Applying
M3.03	Identify the working of a standard fuel cell	3	Applying
CO4	Identify the working of a micro Hydro Turbine Generator		
M4.01	Demonstrate the components of a micro Hydro Turbine Generator	3	Understanding
M4.02	Demonstrate the working of a micro hydro turbine generator	3	Understanding
M4.03	Develop the working model of a standard micro Hydro Turbine Generator unit	3	Applying
M4.04	Draw the water head versus power curve for a given micro hydro power unit.	3	Applying
	Lab test -2	3	

Text / Reference

T/R	Book Title/Author
T1	G.D Rai - Non conventional Energy Sources, Khanna Publishers
T2	Khan B H-Non conventional Energy Resources, Tata McGraw Hill, 2009
R1	Sawhney .G.S Non Conventional Energy Resources, PHI Learning, 2012

Online Resources

Sl.No	Website Link
1	https://eternaluniversity.edu.in/docs/RenewableEnergyandGreenTechnology.pdf
2	http://www.jnkvv.org/PDF/30042020142936304201343.pdf
3	https://www.electricelectronicsindia.com/renewable-energy-engineering-lab-equipment
4	https://www.nvistech.com/technical-training/renewable-energy-training-solutions/experimentation-with-wind-energy